INDUSTRIAL BOILER SAFETY AND OPERATION MANUAL

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1. Purpose and Scope

This manual provides mandatory safety procedures, operating limits, and emergency protocols for all high-pressure steam boilers installed at the facility. It applies to operators, maintenance technicians, and shift supervisors.

2. Operating Pressure Limits

- Maximum Allowable Working Pressure (MAWP): **16 bar**
- Normal operating pressure range: **10 – 12 bar**
- Safety valve set pressure: **15.5 bar**
- Low water cut-off must be tested daily before start-up.

3. Pre-Start Checks

Before lighting the burner, the operator must verify:

1. Water level is visible between 1/3 and 2/3 in the sight glass.
2. All pressure gauges are within calibration date.
3. Forced draft fan is running and air pressure is adequate.
4. Fuel supply valves are open and no leaks are present.
5. Blowdown valves are closed and drain lines are clear.

4. Start-up Procedure

Open the main steam stop valve slowly to avoid water hammer. Ignite the burner only after confirming combustion air is established. Never leave the boiler unattended during the first 30 minutes of operation. Record pressure, water level, and flame condition every 15 minutes during start-up.

5. Emergency Shutdown Procedure

In case of high pressure, low water, flame failure, or any abnormal condition:

1. Press the red Emergency Stop button immediately.
2. Close the main fuel supply valve.
3. If water level is high, open the bottom blowdown valve carefully.
4. Notify the shift supervisor and record the incident.
5. Do not attempt to restart until the cause has been identified and corrected.

6. Daily and Weekly Checks

Daily: Water level, pressure gauge reading, flame appearance, unusual noise or vibration.

Weekly: Test low-water cut-off, inspect safety valve for leakage, check feedwater pump operation.

Monthly: Inspect refractory, clean sight glass, verify calibration of all gauges.

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